Introduction to Machine Learning

1. What is Machine Learning?

Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

2. Artificial Intelligence and Deep Learning

Artificial intelligence (AI) is the broader field of building systems that perform tasks associated with human intelligence. Deep learning is a type of machine learning that uses neural networks with multiple layers to learn complex patterns.

3. Supervised Learning

Supervised learning uses labelled examples. Each training example includes input data and the correct output. Classification predicts categories, while regression predicts numerical values.

4. Unsupervised Learning

Unsupervised learning finds patterns in data without labelled answers. Clustering is a common technique that groups similar data points together.

5. Neural Networks

A neural network is a machine learning model made of connected units called neurons. These units process information in layers and adjust their parameters during training.

6. Training Data and Models

Training data is used to teach a model. The model learns relationships in the data, and then it can make predictions on new examples. Good data quality is important for reliable results.

7. Evaluation

A model should be evaluated on data it has not seen during training. Accuracy can be used for classification, while metrics such as mean squared error can be used for regression.